

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 29)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

In the 1950s a chemical factory at Minamata, Japan, discharged inorganic mercury compounds into a shallow bay. Thousands of local people who ate fish and shellfish from the bay suffered severe neurological damage and many babies were born with birth abnormalities.

0 1

. 1

Explain how the low-toxicity inorganic mercury discharged into the bay came to cause such severe poisoning.

[4 marks]

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0 1

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Explain why methyl mercury caused both neurological damage in adults and birth abnormalities.

[3 marks]

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0 1

. 3

Compare the toxicity of the three chemical forms of mercury.

[3 marks]

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10

0 2 . 1

Describe the main features of the Montreal Protocol (1987).

[3 marks]

0 2 . 2

For each of the following CFC uses, give the alternative process or material now used and one disadvantage of the alternative.

[4 marks]

Aerosol propellant

Refrigerant

0 2 . 3

Describe how waste CFCs from old refrigerators are disposed of safely.

[2 marks]

0	2	.	4
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State **one** reason why the ozone layer is not expected to recover fully for many decades.

[1 mark]

10

0	3
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A dam and reservoir are to be built on a river to supply water and regulate flow.

03.1

Explain **three** effects the reservoir may have on wildlife in and around the river.

[6 marks]

0	3	.	2
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Explain why decomposition of flooded vegetation in a new reservoir contributes to climate change.

[2 marks]

0 3 . 3

Explain how a river-regulation reservoir reduces both flooding and water shortages downstream.

[2 marks]

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10

0 4 . 1

For each of the following investigations, name the most appropriate statistical test.

[4 marks]

Whether noise level decreases as distance from a road increases

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Whether the numbers of dormice differ between woodlands with different management ..

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Whether mean dissolved oxygen differs between two streams

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Whether crop yield increases with fertiliser application (ranked data)

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0 4 . 2

Explain why the chi-squared test cannot be used for data such as wind speed or plant height.

[1 mark]

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0 4 . 3

A Spearman's rank test on 12 pairs of data gave $r_s = -0.72$. The critical value at $p = 0.05$ for $n = 12$ is 0.59. Explain what can be concluded.

[3 marks]

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0 4 . 4

Explain the purpose of error bars on a graph of mean values.

[2 marks]

10

0 5 . 1

Explain the difference between nuclear fission and nuclear fusion.

[2 marks]

0 5 . 2

State the sources of the two fuels used in fusion research.

[2 marks]

1

2

0 5 . 3

Explain why very high temperatures, a vacuum and a magnetic field are needed in a toroidal (torus) fusion reactor.

[3 marks]

0 5 . 4

Give **two** potential advantages of fusion over fission and one reason why it is not yet a commercial energy source.

[3 marks]

10

0 6 . 1

Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

- A Aerobic decomposition produces little methane and returns nutrients ☒ to the soil
- B It produces more electricity ☐
- C It removes heavy metals ☐
- D It destroys asbestos fibres ☐

0 6 . 2

Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

0 6 . 3

Give **two** features of a well-managed landfill site that reduce pollution.

[2 marks]

1

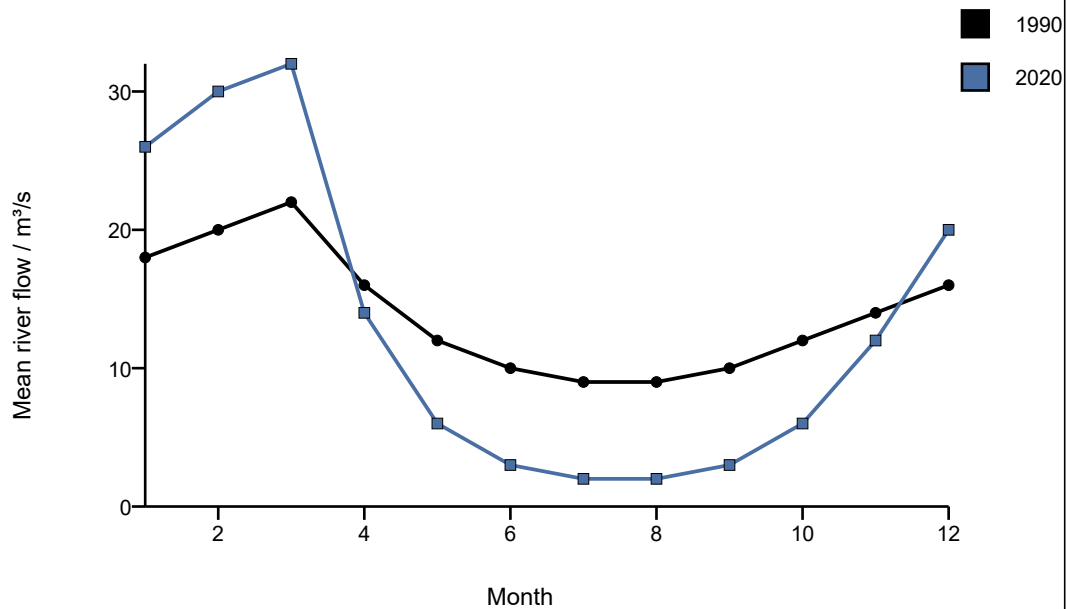
2

5

0 7

Figure 1 shows the monthly flow of a river in South America that is fed partly by glacier meltwater, in 1990 and 2020. The glacier feeding the river shrank considerably between the two dates.

Figure 1



0 7

. 1

Compare the river flow in 1990 and 2020.

[3 marks]

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Explain how the shrinking of the glacier explains the differences.

[3 marks]

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07 . 3 Suggest **two** problems the change in flow causes for people living downstream. **[2 marks]**

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07 . 4 Explain why a glacier can be extending down its valley yet still be losing ice volume. **[2 marks]**

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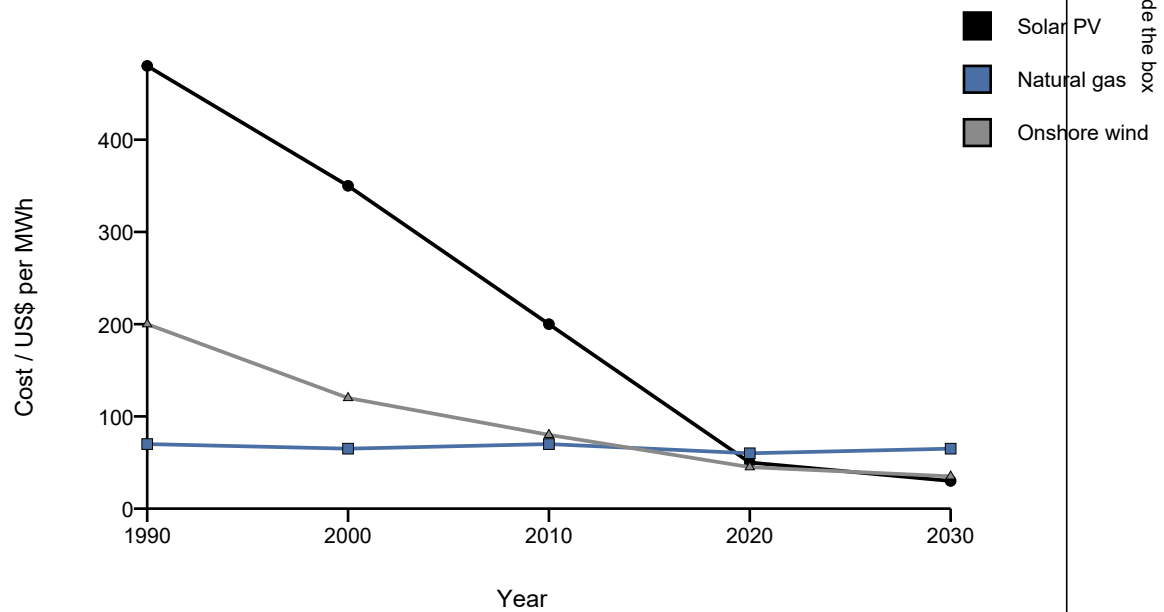
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10

0 8

Figure 1 shows the estimated cost of generating electricity from three sources between 1990 and 2020, and projections to 2030.

Figure 1



0 8

1 Describe the trends shown in Figure 1.

[3 marks]

0 8

2 Explain why new energy technologies are expensive when first introduced and why their costs then fall.

[3 marks]

08 . 3

Explain why a renewable energy project may still be at a financial disadvantage compared with a gas-fired power station of the same total cost.

[2 marks]

08 . 4

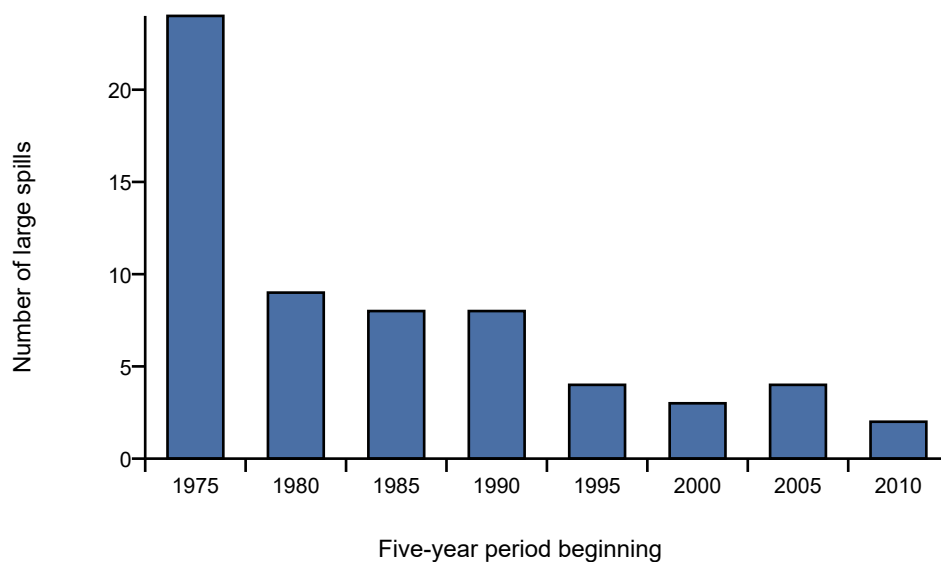
Explain why the price paid for fossil fuel energy does not reflect its full cost.

[2 marks]

10

0 9

Figure 1 shows the number of large oil spills (over 700 tonnes) from tankers worldwide and the amount of oil transported by sea between 1975 and 2015.

Figure 1**Table 1**

Year	1975	1985	1995	2005	2015
Oil transported by sea / million tonnes	1 400	1 200	1 800	2 400	2 900

0 9**1**

Calculate the percentage increase in oil transported by sea between 1975 and 2015.

[2 marks]**Answer**

%**0 9****2**

Describe the trend in the number of large oil spills shown in Figure 1.

[2 marks]

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09 . 3

Suggest **three** reasons for the decline in the number of large spills despite the increase in oil transported.

[3 marks]

1

2

3

09 . 4

Explain why the total amount of oil pollution entering the sea may not have fallen as much as the number of large tanker spills.

[3 marks]

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10 . 1

Evaluate the risks and benefits of the uses of ionising radiation, and explain the principles used to control exposure of workers and the public.

[9 marks]

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[illegible]

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1 0 . 2

Give **one** reason why the risks and benefits of nuclear electricity may not fall on the same people.

[1 mark]

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10

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
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Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

[25 marks]

OR

1 1 . 2

To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

END OF QUESTIONS